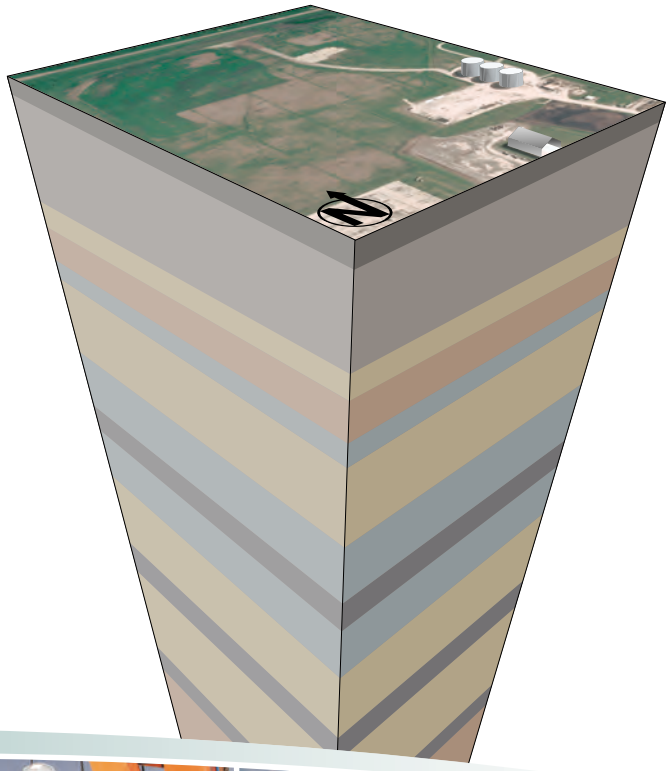
MIDWEST GEOLOGICAL
SEQUESTRATION CONSORTIUM

IBDP

Illinois Basin - Decatur Project

Overview of the Illinois Basin - Decatur Project

The Midwest Geological Sequestration Consortium's (MGSC) Illinois Basin – Decatur Project (IBDP) is a collaboration of the MGSC, the Archer Daniels Midland Company (ADM), Schlumberger Carbon Services, and other subcontractors to inject 1 million metric tons of anthropogenic carbon dioxide (CO_2) into a saline reservoir, the Mount Simon Sandstone, at Decatur, IL. MGSC is one of seven US Department of Energy (DOE) Regional Carbon Sequestration Partnerships and is administered through DOE's National Energy Technology Laboratory. The IBDP derived its CO_2 from ADM's ethanol fermentation plant at Decatur and consists of a compression/dehydration facility, a delivery pipeline, one injection well, one deep observation/verification well, and a geophysical test well, all developed on the ADM-owned site. The project held an Underground Injection Control (UIC) Class I Nonhazardous permit from the Illinois EPA and now holds a UIC Class VI permit issued by the US EPA, Region 5, for post-injection monitoring.









The objectives of the project are to validate the capacity, injectivity, and containment of the Mount Simon, which represents the primary carbon storage resource in the Illinois Basin and the Midwest Region. Operational injection started on November 17, 2011 and was completed successfully on November 26, 2014 with a total volume of 999,215 tonnes.

During the three-year injection phase, extensive data collection occurred. A full subsurface and surface Monitoring, Verification, and Accounting (MVA) program is in place, and periodic data collection such as fluid sampling, geophysical measurements, and cased-hole logging will continue for at least three years post-injection. Core and log data from the original drilling operations have been integrated with a 3D seismic volume to interpret original depositional systems and support reservoir simulation. Overall, capacity, injectivity, and containment have met pre-injection expectations, and project activities remain focused on MVA, understanding CO_2 distribution, and post-injection assessments.

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Project Highlights:

- Permitted Injection Volume: 1 million metric tons
- Target Reservior: Mount Simon Sandstone
- Depth of Reservoir Top: 5,545 ft (1,690.5 m)
- Thickness of Reservoir: 1,506 ft (459.1 m)
- Reservoir Seal: Eau Claire Shale
- Depth to Seal Top: 5,047 ft (1,538.7 m)
- Injection Initiated: November 17, 2011
- Injection Completed: November 26, 2014
- Injection Rate: 1,000 metric tons per day
- UIC Injection Permit Holder: Archer Daniels Midland Company
- Carbon Dioxide Source: Fermentation for ethanol production
- Compression Equipment: Dual four-stage reciprocating with glycol dehydration
- Wellhead Injection Pressure: 1,350 psi (9.3 MPa)
- Wellhead Injection Temperature: 95° F (35° C)
- Delivery Pipeline: 6-inch (15.24 cm) carbon steel, 1.2 mile (1.9 km) length
- Depth of Injection Interval: 6,985–7,050 ft (2,129.6–2,149.4 m)

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